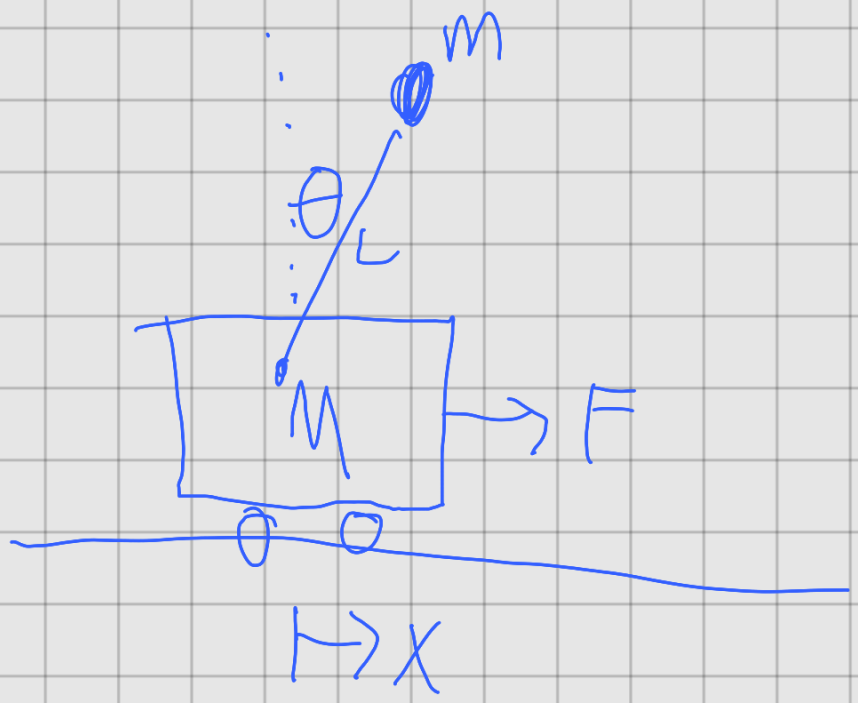
CP4 EOM Derivation by hand.

FBD



$$X = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} \begin{matrix} \text{pos} \\ \text{velocity} \\ \text{angle} \\ \text{angular velocity} \end{matrix}$$

$L = T - V$

T = kinetic

V = potential

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}$$

Cart pos

$x_c = x \quad \dot{x}_c = \dot{x}$

$y_c = 0 \quad \dot{y}_c = 0$

location of pendulum mass

$$x_m = x + l\sin\theta \qquad \dot{x}_m = \dot{x} + l\cos(\theta)\cdot\dot{\theta}$$

$$y_m = l\cos\theta \qquad \dot{y}_m = -l\sin\theta\,\dot{\theta}$$

$$\ddot{x}_m = \ddot{x} + l\left(-\sin(\theta)\cdot\dot{\theta}\cdot\dot{\theta} + \cos(\theta)\ddot{\theta}\right)$$

$$\ddot{x}_m = \ddot{x} - l\sin(\theta)\dot{\theta}^2 + l\cos(\theta)\ddot{\theta}$$

$$\ddot{y}_m = -l\left(\cos(\theta)\dot{\theta}\cdot\dot{\theta} + \sin(\theta)\ddot{\theta}\right)$$

$$\dot{y}_m = -l\cos(\theta)\dot{\theta}^2 - l\sin(\theta)\ddot{\theta}$$

$$T = \frac{1}{2}mv^2$$

$$T_{cart} = \frac{1}{2}M\dot{x}^2$$

$$T_{pen} = \frac{1}{2}m\left((\dot{x}_m)^2 + (\dot{y}_m)^2\right)$$

$$= \frac{1}{2}m\left((\dot{x} + l\cos(\theta)\dot{\theta})^2 + (-l\sin(\theta)\cdot\dot{\theta})^2\right)$$

$$= \frac{1}{2}m\left(\dot{x}^2 + 2l\cos(\theta)\dot{\theta} + l^2\cos^2(\theta)\dot{\theta}^2 + l^2\sin^2(\theta)\dot{\theta}^2\right)$$

$$+ \qquad = 1$$

$$T_{pen} = \frac{1}{2}m\left(\dot{x}^2 + 2l\cos(\theta)\dot{\theta} + l^2\dot{\theta}^2\right)$$

$$T = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left(\dot{x}^2 + 2\dot{x}l\cos(\theta)\dot{\theta} + l^2\dot{\theta}^2\right)$$

$$T = \frac{1}{2}(M+m)\dot{x}^2 + \dot{x}ml\cos(\theta)\dot{\theta} + \frac{1}{2}ml^2\dot{\theta}^2$$

<u>$V = Pot$</u>

$$V_{cart} = 0$$

$$V_{pen} = m\cdot g\cdot y_m = mgl\cos\theta$$

$$L = T - V$$

$$L = \frac{1}{2}(M+m)\dot{x}^2 + \dot{x}ml\cos(\theta)\dot{\theta} + \frac{1}{2}ml^2\dot{\theta}^2 - mgl\cos\theta$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) = \frac{\partial L}{\partial q_j} \qquad q_j = x, \theta$$

$$\frac{\partial L}{\partial \dot{x}} = (M+m)\dot{x} + m l \cos(\theta)\dot{\theta}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = (M+m)\ddot{x} + \left(-ml\sin(\theta)\dot{\theta}\dot{\theta} + ml\cos(\theta)\ddot{\theta}\right)$$

$$\frac{\partial L}{\partial x} = 0$$

eqn 1

$$(M+m)\ddot{x} - ml\sin(\theta)\dot{\theta}^2 + m l\cos(\theta)\ddot{\theta} = F - b\dot{x}$$

$$\frac{\partial L}{\partial \dot{\theta}} = \dot{x}\, ml\cos(\theta) + m l^2 \dot{\theta}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = ml\frac{d}{dt}\left(\cos(\theta)\dot{x}\right) + ml^2\ddot{\theta}$$

$$= ml\left(-\sin(\theta)\dot{\theta}\dot{x} + \cos\theta\,\ddot{x}\right) + ml^2\ddot{\theta}$$

$$= ml^2\ddot{\theta} - ml\sin(\theta)\dot{x}\dot{\theta} + ml\cos(\theta)\ddot{x}$$

$$\frac{\partial L}{\partial \theta} = -\dot{x}\, ml\sin(\theta)\dot{\theta} + mgl\sin(\theta)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$$

eqn 2

$$ml^2\ddot{\theta} - \cancel{ml\sin(\theta)\dot{x}\dot{\theta}} + ml\cos(\theta)\ddot{x} - mgl\sin(\theta) + \cancel{\dot{x}\, ml\sin(\theta)\dot{\theta}} = 0$$

$$ml^2\ddot{\theta} + ml\cos(\theta)\ddot{x} - mgl\sin(\theta) = 0$$

EOM

$$(M+m)\ddot{x} + ml\cos(\theta)\ddot{\theta} - ml\sin(\theta)\dot{\theta}^2 - F + b\dot{x} = 0$$

$$ml^2\ddot{\theta} + ml\cos(\theta)\ddot{x} - mgl\sin(\theta) = 0$$

state space form

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} (M+m) & ml\cos(\theta) \\ ml\cos\theta & ml^2 \end{bmatrix}}_{M(q)} \underbrace{\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix}}_{\ddot{q}} = \underbrace{\begin{bmatrix} ml\sin(\theta)\dot{\theta}^2 + F - b\dot{x} \\ mgl\sin\theta \end{bmatrix}}_{f(q,\dot{q},u)}$$

$$M(q)^{-1} = \frac{1}{\det}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det = \left((M+m)ml^2\right) - \left(m^2l^2\cos^2(\theta)\right)$$

$$Mml^2 + m^2l^2 - m^2l^2\cos^2\theta$$

$$ml^2\left(M + m - m\cos^2(\theta)\right)$$

$$M(q)^{-1} = \frac{1}{ml^2(M+m-m\cos^2(\theta))}\begin{bmatrix} ml^2 & -ml\cos\theta \\ -ml\cos\theta & M+m \end{bmatrix}$$

$$\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = M(q)^{-1} \cdot \begin{bmatrix} ml\sin(\theta)\dot{\theta}^2 + F - b\dot{x} \\ mgl\sin\theta \end{bmatrix}$$

$$\ddot{x} = \frac{1}{ml^2(M+m-m\cos^2(\theta)}\left(ml^2\left(ml\sin(\theta)\dot{\theta}^2 + F - b\dot{x}\right) - ml\cos\theta \cdot mgl\sin\theta\right)$$

$$\ddot{\theta} = \frac{1}{\det}\left(-ml\cos(\theta)\left(ml\sin(\theta)\dot{\theta}^2 + F - b\dot{x}\right) + (M+m)mgl\sin\theta\right)$$

state space

$$\det = (M+m)ml^2 - m^2l^2\cos^2 x_3$$

$$\dot{x} = \begin{bmatrix} x_2 \\ \frac{1}{\det}\left(m^2 l^3 \sin(x_3)\, x_4^2 + m l^2 F - m l^2 b x_2 + m^2 l^2 g \cos x_3 \sin x_3\right) \\ x_4 \\ \frac{1}{\det}\left((M+m)\, m g l \sin x_3 - m^2 l^2 \cos(x_3)\sin(x_3)\, x_4^2 - m l \cos(x_3) F + m l \cos(x_3) b x_2\right) \end{bmatrix}$$

linearized about $\theta = 0$ $\quad \sin\theta = \theta \quad \cos\theta = 1 \quad \dot{\theta} = 0$

$$\det L = (M+m)\, m l^2 - m^2 l^2$$

$$\dot{x} = \begin{bmatrix} x_2 \\ \frac{1}{\det L}\left(m l^2 F - m l^2 b x^2 - m^2 l^2 g\, x_3\right) \\ x_4 \\ \frac{1}{\det L}\left((M+m)\, m g l\, x_3 - m l F + m l b x_2\right) \end{bmatrix}$$

$$\dot{x} = Ax + Bu$$

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{mlb}{\det L} & -\frac{m^2 l^2 g}{\det L} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{mlb}{\det L} & \frac{(M+m)\, m g l}{\det L} & 0 \end{bmatrix}}_{A} x + \underbrace{\begin{bmatrix} 0 \\ \frac{m l^2}{\det L} \\ 0 \\ \frac{-m l}{\det L} \end{bmatrix}}_{B} F$$

LQR $\quad Q = \mathrm{diag}\left[\overset{\text{der } x}{5}, \overset{\text{der } \dot{x}}{1}, \overset{\text{der } \theta}{100}, \overset{\text{der } \dot{\theta}}{10}\right]$

main error we want to prevent

$R = [1]$